

		FAILURE MODE AND EFFECTS ANALYSIS (PROCESS FMEA)										Format No : F/NPD/07							
		Project name:-		952NX		Component Part No.: 92187158		PFMEA no.:MPLUPFPPCHFFPFMEA/05050187158		Released on : 01/07/2022									
		Team Leader:		Mr. Satish Pandey		Component Description-DOOR, ASSY, LH, 952NX				Revised on : 00									
		Team Members:		Mr. M.S/SPRSB/DD/VNKA/AD/OM/SS/AJ		Component Revision No.:- C		PFMEA Last Review Date : 17-07-2025		Revision No : 00									
PROTOTYPE				PRE-LAUNCH				PRODUCTION											
No.	Process Function	Potential Failure Mode	Potential Effect(s) of Failure	Severity	CL	SS	Potential Cause(s) of Failure	Occurrence	Current Controls Prevention	Current Controls Detection	Detection	RPN	Recommended Action	Responsibility & Completion Date	Action Taken	Revised			
																Sev	Occ	Det	RPN
SUB-ASSY -92187146-1 -Door WA,LH,952NX, Rev-C																			
10	Child Part Collection	Wrong child part used	M/C: Part may be not fitted SO: Operation. Effected CU: Flimnet issue EU: NIL				Wrong part taken		Part no. wise child part storage	Failure Control		0							
20	Tack Welding	Visual Appearance	NO: Part visual appearance poor (4) SO: No Effect (1)				1. Weld feed rate high 2. Oil surface 3. Untrained Operator		1. Set up approval by attribute method	1. Operator Training for Welding		0							
		Weld Strength	NO: Part failure during testing, may be complete lot segregate (6) SO: No Effect (1) CE: Customer high dis-satisfied may be reject the			1. Process parameter variation 2. Oil surface 3. Less Weld Area 4. Fan Over Head		1. Set up approval by attribute method	1. SET UP INSPECTION OF PART 2. INPROCESS INSPECTION WITH BEND GAUGE		0								
30	Full Welding	Visual Appearance	NO: Part visual appearance poor (4) SO: No Effect (1)				1. Weld feed rate high 2. Oil surface 3. Untrained Operator		1. Set up approval by attribute method	1. Operator Training for Welding		0							
		Weld Strength	NO: Part failure during testing, may be complete lot segregate (6) SO: No Effect (1) CE: Customer high dis-satisfied may be reject the			1. Process parameter variation 2. Oil surface 3. Less Weld Area 4. Fan Over Head		1. Set up approval by attribute method	1. SET UP INSPECTION OF PART 2. INPROCESS INSPECTION WITH BEND GAUGE		0								
40	Finishing/Deburring	GRINDING MARKS ON THE SURFACE	M/C: NIL SO: PART MAY BE REJECTED CU: NIL EU: APPEARANCE NOT SMOOTH				1. LOW GRIT GRINDING WHEEL USED		1. USE HIGH GRIT GRINDING WHEEL	1. 100% VISUAL INSPECTION BY OPERATOR		0							
50	Move for Next Process																		
SUB-ASSY -92187146-2 -Door WA,LH,952NX, Rev-C																			
10	Child Part Collection	Wrong child part used	M/C: Part may be not fitted SO: Operation. Effected CU: Flimnet issue EU: NIL				Wrong part taken		Part no. wise child part storage	Failure Control		0							
20	Tack Welding	Visual Appearance	NO: Part visual appearance poor (4) SO: No Effect (1)				1. Weld feed rate high 2. Oil surface 3. Untrained Operator		1. Set up approval by attribute method	1. Operator Training for Welding		0							
		Weld Strength	NO: Part failure during testing, may be complete lot segregate (6) SO: No Effect (1) CE: Customer high dis-satisfied may be reject the			1. Process parameter variation 2. Oil surface 3. Less Weld Area 4. Fan Over Head		1. Set up approval by attribute method	1. SET UP INSPECTION OF PART 2. INPROCESS INSPECTION WITH BEND GAUGE		0								
30	Full Welding	Visual Appearance	NO: Part visual appearance poor (4) SO: No Effect (1)				1. Weld feed rate high 2. Oil surface 3. Untrained Operator		1. Set up approval by attribute method	1. Operator Training for Welding		0							
		Weld Strength	NO: Part failure during testing, may be complete lot segregate (6) SO: No Effect (1) CE: Customer high dis-satisfied may be reject the			1. Process parameter variation 2. Oil surface 3. Less Weld Area 4. Fan Over Head		1. Set up approval by attribute method	1. SET UP INSPECTION OF PART 2. INPROCESS INSPECTION WITH BEND GAUGE		0								
40	Finishing/Deburring	GRINDING MARKS ON THE SURFACE	M/C: NIL SO: PART MAY BE REJECTED CU: NIL EU: APPEARANCE NOT SMOOTH				1. LOW GRIT GRINDING WHEEL USED		1. USE HIGH GRIT GRINDING WHEEL	1. 100% VISUAL INSPECTION BY OPERATOR		0							
50	Move for Next Process																		
Full Welding -92187146-Door WA,LH,952NX, Rev-C																			
10	Child Part Collection	Wrong child part used	M/C: Part may be not fitted SO: Operation. Effected CU: Flimnet issue EU: NIL				Wrong part taken		Part no. wise child part storage	Failure Control		0							
20	Full Welding	Visual Appearance	NO: Part visual appearance poor (4) SO: No Effect (1)				1. Weld feed rate high 2. Oil surface 3. Untrained Operator		1. Set up approval by attribute method	1. Operator Training for Welding		0							
		Weld Strength	NO: Part failure during testing, may be complete lot segregate (6) SO: No Effect (1) CE: Customer high dis-satisfied may be reject the			1. Process parameter variation 2. Oil surface 3. Less Weld Area 4. Fan Over Head		1. Set up approval by attribute method	1. SET UP INSPECTION OF PART 2. INPROCESS INSPECTION WITH BEND GAUGE		0								
30	Finishing/Deburring	GRINDING MARKS ON THE SURFACE	M/C: NIL SO: PART MAY BE REJECTED CU: NIL EU: APPEARANCE NOT SMOOTH				1. LOW GRIT GRINDING WHEEL USED		1. USE HIGH GRIT GRINDING WHEEL	1. 100% VISUAL INSPECTION BY OPERATOR		0							
40	Move for Next Process																		
Painting (DOOR, ASSY, LH, 952NX-92244728) (Power Tan -Std-86628042)																			
10	Transportation to pretreatment	Material Spillage, wrong placement	Material Loss and Damage				Material Overflow		SDP	Area wise SS audit & CLR check sheet As per production plan, and PQ.		0							
20	Knock of Degrease	Total alkali less than 70 ml	Surface not cleaned properly				1. Wrong stacking 2. Un Skilled Transport operator LR Crane		1. Stack Designed 2. 100% visual insp. by operator Periodic insp. by inspector	Inspection of the Titration once in every shift before start of the process		0							
		Temperature less than 50°C	Surface not cleaned properly			Due to continuous production Bondrite AK-5055 is less		1. Addition of the ARVN water. 2. Heater not working.	Inspection of the Titration once in every shift before start of the process.		0								
30	Degrease	Pressure of water less than 0.8 kg/Cm2	Part cleaning not done properly				More Bondrite AK - 5055 is mixed by operator		1. Visual inspection by the operator in pressure gauge before start of the process and 2 times visually in every shift	Inspection of the Titration once in every shift before start of the process		0							
		Oil content more than 40g/Ltr	Part surface should be oily			Filter bag choke due to dust		2. Daily checking of the temperature before start of the process. 2. Controller stop the process automatically if temperature setting down.	Inspection of the Titration once in every shift before start of the process		0								
40	Water Rinse 1	Total alkali less than 90 ml	Surface not cleaned properly				1. Oil separator off 2. Oil separator not working properly		Using oil separator to remove oil from water	Tank water inspection in third party lab once in every month		0							
		Temperature less than 50°C	Surface not cleaned properly			Due to continuous production Bondrite AK-5055 is less		1. Inspection of the Titration once in every shift before start of the process. 2. Draining given to operator for material misup	Inspection of the Titration once in every shift before start of the process		0								
50	Water Rinse 2	Pressure of water less than 0.8 kg/Cm2	Part cleaning not done properly				Filter bag choke due to dust		1. Visual inspection by the operator in pressure gauge before start of the process and 2 times visually in every shift	Inspection of the Titration once in every shift before start of the process		0							
		Oil content more than 40g/Ltr	Part surface should be oily			1. Addition of the ARVN water. 2. Heater not working.		2. Daily checking of the temperature before start of the process. 2. Controller stop the process automatically if temperature setting down.	Inspection of the Titration once in every shift before start of the process		0								
60	Surface Activation	Removal of alkali content	Customer dissatisfaction due to appearance not ok				1. Oil separator off 2. Oil separator not working properly		Using oil separator to remove oil from water	Tank water inspection in third party lab once in every month		0							
70	Phosphate	Uneven phosphate layer on job	Customer dissatisfaction due to appearance not ok				1. Visual inspection by the operator in pressure gauge before start of the process and 2 times visually in every shift		1. Inspection of the Titration once in every shift before start of the process. 2. Draining given to operator for material misup	Inspection of the Titration once in every shift before start of the process		0							
80	Water Rinse 03	Traces of acid	Customer dissatisfaction due to appearance not ok				1. Addition of the ARVN water. 2. Heater not working.		2. Daily checking of the temperature before start of the process. 2. Controller stop the process automatically if temperature setting down.	Inspection of the Titration once in every shift before start of the process		0							
90	Water Rinse 04	Traces of acid	Customer dissatisfaction due to appearance not ok				Filter bag choke due to dust		1. Visual inspection by the operator in pressure gauge before start of the process and 2 times visually in every shift	Inspection of the Titration once in every shift before start of the process		0							
100	PTPCDM	Cleaning of component not ok	Customer dissatisfaction due to appearance not ok				1. Oil separator off 2. Oil separator not working properly		Using oil separator to remove oil from water	Tank water inspection in third party lab once in every month		0							
110	CEC Coating	Dust, rughair, crater, lump,grit/sand,uncover,scratches etc.	Customer dissatisfaction due to appearance not ok				1. Visual inspection by the operator in pressure gauge before start of the process and 2 times visually in every shift		1. Inspection of the Titration once in every shift before start of the process. 2. Draining given to operator for material misup	Inspection of the Titration once in every shift before start of the process		0							
120	Ultra Filtration	Removal of Excess paint not done properly	Customer dissatisfaction due to appearance not ok				1. Addition of the ARVN water. 2. Heater not working.		2. Daily checking of the temperature before start of the process. 2. Controller stop the process automatically if temperature setting down.	Inspection of the Titration once in every shift before start of the process		0							
130	IDRCDM	Removal of Excess paint not done	Customer dissatisfaction due to appearance not ok				Filter bag choke due to dust		1. Visual inspection by the operator in pressure gauge before start of the process and 2 times visually in every shift	Inspection of the Titration once in every shift before start of the process		0							
140	Unloading	Dust, rughair,craters, lump,grit/sand,uncover,scratches etc.	Customer dissatisfaction due to appearance not ok				1. Oil separator off 2. Oil separator not working properly		Using oil separator to remove oil from water	Tank water inspection in third party lab once in every month		0							
150	Baking	Temperature is less/more than required	Customer dissatisfaction due to appearance not ok				1. Visual inspection by the operator in pressure gauge before start of the process and 2 times visually in every shift		1. Inspection of the Titration once in every shift before start of the process. 2. Draining given to operator for material misup	Inspection of the Titration once in every shift before start of the process		0							



Released on : 01/07/2022

Revision No. : 00

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PFMEA No.: MAPLPPW/CNH/PFMEA
050/92187158

PFMEA Last Review Date : 17-07-2025

PROTOTYPE

PRE-LAUNCH

PRODUCTION

No.	Process Function	Potential Failure Mode	Potential Effect(s) of Failure	Severity	CLASS	Potential Cause(s) of Failure	Current Controls Prevention	Current Controls Detection	RPN	Recommended Action	Responsibility & Completion Date	Action Taken	Revised		
													Sev.	Occ.	Det.
Sheet Metal Parts PFMEA															
10	RM Procurement	MATERIAL FROM APPROVED SOURCE	MCC-NIL SD - Operation Effective CU - Part Failure EU - MCC Breakdown			Material from local market	Material from Approved source	Approved Source supplier list							
		* THICKNESS OVER SIZE	MCC-NIL SD - Banding Dim. Effective CU - Finest Issue EU - NIL			1. Wrong PO / incomplete Details in PO issued to supplier	1. Verification by Sign. Authority of PO with Purchase indent and before issue to supplier.	1. MATERIAL CROSS CHECK BY QA INSPECTOR BEFORE ENTERING THE PLANT AGAINST REC A JUDGE.							
						2. Wrong PO / incomplete Details in PO issued to supplier	2. Check at receiving stage	2. MR TC should be verified by field inspector							
						3. No verification of TC by QA inspection	3. Third party material test Plan to be adhered	3. Verification of MR TC material by 90% and third party test Report.							
20	INCOMING INSPECTION	* THICKNESS UNDER SIZE	MCC-NIL SD - Banding Dim. Effective CU - Finest Issue EU - NIL			4. NON CALIBRATED INSTRUMENT USE FOR INSPECTION	4. Instrument calibration plan to be adhered	4. Calibrated instrument should be used to check the material.							
						1. Wrong PO / incomplete Details in PO issued to supplier	1. Verification by Sign. Authority of PO with Purchase indent and before issue to supplier.	1. MATERIAL CROSS CHECK BY QA INSPECTOR BEFORE ENTERING THE PLANT AGAINST REC A JUDGE.							
						2. Wrong TC Provided by Supplier.	2. Check at receiving stage	2. MR TC should be verified by field inspector							
						3. No verification of TC by QA inspection	3. Third party material test Plan to be adhered	3. Verification of MR TC material by 90% and third party test Report.							
30	Storage	* PLATE BUSTY	MCC-NIL SD - Poor surface Treatment CU - Fall in line EU - Durability Effective			1. Material Stored in open condition	1. Material Stored in covered area. Awareness about it to Quality inspection	1. Incoming Inspection Report. Before next process operator will ensure that Plate is free from RUST.							
						2. FFCO not maintained	2. FFCO system applied	2. Color coding to be done on the steel plates. Green, Yellow.							
		* SHEET BEND	MCC-NIL SD - Banding Dim Effective CU - Finest Issue EU - Durability Effective			1. Plate may be hit / damage during unloading	1. PROPER LOADING & UNLOADING BY HSDT (GRABE)	1. Incoming Inspection Report							
						2. Plate may not be stored properly	2. STORAGE LOCATION DEFINED FOR RAW MATERIAL	2. Before next process start operator should ensure that Plate is not bent / Improves inspection Report.							
40	Laser Cutting	* MATERIAL GRADE WRONG	MCC-NIL SD - Banding Effective CU - Part Check EU - Durability Effective			1. Wrong PO / incomplete Details in PO issued to supplier	1. Verification by Sign. Authority of PO with Purchase indent and before issue to supplier.	1. Incoming Inspection Report							
						2. Wrong TC Provided by Supplier.	2. Verification of TC by field inspector.	2. Third party test to be done as per plan.							
						3. Chemical and Physical properties not matched with spec.	3. Chemical and Physical properties be checked & matched with spec.	3. Incoming Inspection Report							
						4. No proper place for storage of material by grade and thickness	4. Storage area is identified as per grade and thickness spec.	4. W/ Displayed near that material Storage area and Incoming Inspector Green colour code after inspection.							
50	Oxy Cutting	* TAPPER CUTTING	MCC-NIL SD - Part may reject. Excess rework CU - Finest issue EU - NIL			1. MR parameter may not be set as per job.	1. MR process parameter area to be set as per job, before process starts	1. Incoming Inspection Report with Full Size Template. 2. Daily Process parameter check sheet.							
						2. Bed may not be clean.	2. Bed cleaning to be done as per decided frequency	2. Bed cleaning date to be mention on bed.							
		* PROFILE OVER SIZE	MCC-NIL SD - Effect on bend operating CU - Finest issue EU - Durability Effective			1. Part Heating is not ok.	1. Verification of heating of machine before process starts.	1. 1st set up inspection of part with Full Size Template.							
						2. Process parameter of rnc is not ok as per Std.	2. Reset process parameter of plasma rnc	2. Incoming Inspection Report with Full Size Template.							
60	Drilling	* PROFILE UNDER SIZE	MCC-NIL SD - Part reject CU - Finest issue EU - NIL			1. Part Heating is not ok	1. Verifications of heating of machine before process starts	1. 1st set up inspection of part with Full Size Template.							
						2. Process parameter of rnc is not ok as per Std.	2. Reset process parameter of plasma rnc	2. Incoming Inspection Report with Full Size Template.							
		* UNDEVEN CUTTING / SLAG	MCC-NIL SD - Effect on Bend operation CU - Finest issue EU - NIL			Process parameter of rnc is not ok as per Std.	Reset process parameter of plasma rnc	1. 1st set up inspection of part 2. Incoming Inspection Report with Full Size Template							
70	Tapping	* TAPPER CUTTING	MCC-NIL SD - Part may reject. Excess rework CU - Finest issue EU - NIL			1. MR parameter may not be set as per job.	1. MR process parameter area to be set as per job, before process starts	1. Incoming Inspection Report with Full Size Template. 2. Daily Process parameter check sheet.							
						2. Bed may not be clean.	2. Bed cleaning to be done as per decided frequency	2. Bed cleaning date to be mention on bed.							
		* PROFILE OVER SIZE	MCC-NIL SD - Effect on bend operating CU - Finest issue EU - Durability Effective			1. Part Heating is not ok.	1. Verification of heating of machine before process starts.	1. 1st set up inspection of part with Full Size Template.							
						2. Process parameter of rnc is not ok as per Std.	2. Reset process parameter of plasma rnc	2. Incoming Inspection Report with Full Size Template.							
80	Chamfering	* UNDEVEN CUTTING / SLAG	MCC-NIL SD - Effect on Bend operation CU - Finest issue EU - NIL			Process parameter of rnc is not ok as per Std.	Reset process parameter of plasma rnc	1. 1st set up inspection of part 2. Incoming Inspection Report with Full Size Template							
90	Straightening	DRILL UNDER OR OVER SIZE	MCC-NIL SD - Part reject CU - Finest issue EU - NIL			WRONG SIZE DRILL SELECTION	DRILL SIZE TO BE CHECK BEFORE USE	DIMENSIONS ARE CHECKED BY OPERATOR AFTER COMPLETING OPERATION.							
100	Bending/Forming	TAPPING THREAD DEPTH MAY NOT BE CORRECT	MCC-NIL SD - Part may reject CU - Finest issue EU - NIL			ALL 63 TAP NOT RUN IN TAPPING HOLE.	TAPPING HOLE TO BE CHECK AFTER OPERATION COMPLETE. THROUGH GO NO GO GAUGE	TAPPING HOLE CHECKED BY GO NO GO GAUGE.							
110	Storage	* TAPPING THREAD DEPTH MAY NOT BE CORRECT	MCC-NIL SD - Part may reject CU - Finest issue EU - NIL			ALL 63 TAP NOT RUN IN TAPPING HOLE.	TAPPING HOLE TO BE CHECK AFTER OPERATION COMPLETE. THROUGH GO NO GO GAUGE	TAPPING HOLE CHECKED BY GO NO GO GAUGE.							

120	Move for Next Process																			
ABBREVIATION:-		M/C- MACHINE	SO-SECOND OPERATION		CU- CUSTOMER USE		EU:- EXTERNAL USE													
Prepared By : Mr.Amit Vishwakarma										Approved By- Mr.Prashant Raj Srivastava										
DATE		REV NO.		CHANGES										Prepared By		Approved By				
17-Jul-25		A		Initial for revision no. "C"										Mr.Amit Vishwakarma		Mr. Prashant Raj Srivastava				
Note		PFMEA UPDATE Frequency : During customer complaint / Any process change / Yearly review																		
		ACTION PLAN RULE : Action shall we taken if any RPN value crossed over 100 otherwise no need to take any action																		

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